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Classification of Tea Characteristics Elements Based on Laser-Induced Breakdown Spectroscopy and Modeling of Fermentation

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Abstract: To achieve the rapid identification of tea varieties and the precise quantitative assessment of fermentation quality, this study investigates tea classification and fermentation quality prediction using laser-induced breakdown spectroscopy (LIBS) combined with variable importance in projection (VIP) feature selection and the Random Forest (RF) algorithm. Thirty-eight varieties of green and oolong teas were used as research subjects. LIBS spectral data in the 181–835 nm range were collected. The VIP algorithm was utilized to extract highly correlated characteristic elements corresponding to spectral peaks. Based on the optimal feature parameters, an RF classification model was constructed. Coupled with a Pearson correlation coefficient threshold discrimination strategy, this approach achieved 100% accurate discrimination between major categories (green and oolong teas), distinct batches of the same variety, and pure versus mixed samples. Furthermore, quantitative modeling was conducted for the core fermentation quality indicators (moisture, water extracts, and tea polyphenols) of Pu-erh tea. The results demonstrate that the modeling method coupling VIP feature optimization with RF can effectively mine the spectral elemental response patterns corresponding to fermentation differences, thereby increasing the identification accuracy of tea varieties and batches to 100%. This study provides reliable technical support for the intelligent, non-destructive testing of tea and the standardized optimization of fermentation processes, presenting promising prospects for practical application.

Keywords: Laser-induced breakdown spectroscopy (LIBS); VIP feature selection; Tea classification; Fermentation quality

1. Introduction

As one of the world’s three major non-alcoholic beverages, the quality and commercial value of tea are primarily determined by tea plant cultivars, growth environments and processing fermentation techniques[1]. China possesses rich tea resources and a highly complex classification system. However, issues such as cultivar confusion, batch mixing, and adulteration frequently occur during tea production, processing, and market trading, severely hindering the standardized development of the tea market[2]. Fermentation is the core process that dictates the quality and flavor profile of tea. Green tea, oolong tea, and Pu-erh tea correspond to typical non-fermented, semi-fermented, and post-fermented processing methods, respectively. Gradient variations in fermentation degree directly alter the composition and concentration of core internal substances, such as tea polyphenols, moisture, and organic matter, ultimately forming differentiated quality characteristics

and unique elemental spectral fingerprints. Therefore, establishing an accurate and rapid method for classifying tea fermentation attributes and achieving quantitative detection of typical tea fermentation quality holds significant theoretical importance and practical application value for precise quality control and fermentation process optimization.

At present, tea quality identification and detection primarily rely on sensory evaluation and traditional physicochemical analysis. Sensory evaluation depends on the subjective judgment of professional tea tasters based on sight, smell, and taste. This method suffers from high subjectivity, poor reproducibility, and a lack of standardization, making it inadequate for rapid, large-batch tea testing[3]. Traditional physicochemical detection requires complex sample preprocessing, such as grinding and extraction. Traditional physicochemical detection methods necessitate complex sample preprocessing, such as grinding and extraction. These procedures are cumbersome, time-consuming, and highly destructive, failing to meet the industrial demands for the non-destructive and efficient evaluation of tea. Conversely, optical spectroscopic detection technology, owing to its advantages of rapid, non-destructive, high-precision, and synchronous multi-parameter analysis, has been widely applied in the quality discrimination and quantitative compositional analysis of agricultural products. This provides a novel technical pathway for the intelligent detection of tea[4-5].

Laser-induced breakdown spectroscopy (LIBS) enables rapid acquisition of elemental composition and structural information of samples, thereby sensitively tracing subtle quality variations of tea originating from cultivar discrepancy, manufacturing procedure and fermentation degree[6,7]. Xu et al.[8] combined LIBS with principal component analysis (PCA) to discriminate three tea categories (black, green and white tea) with 100% overall accuracy, and the average classification accuracy for sub-varieties within each tea group reached 98.3%. Zhang et al.[6] utilized LIBS coupled with partial least-squares discriminant analysis (PLS-DA) to identify six tea varieties including green tea, golden-tip black tea and Tieguanyin, achieving an accuracy of 99.77%. Jiang[9] adopted hyperspectral spectroscopy integrated with support vector machine (SVM) to grade mechanically processed Longjing green tea, yielding 98% identification accuracy. Ma et al.[10] proposed a multi-energy LIBS strategy combined with a Northern Goshawk optimizer-particle swarm optimized random forest (NGO-PSO-RF) model, which attained a test-set accuracy of 99.22% for eight tea species. Nevertheless, most existing studies only concentrate on coarse inter-category discrimination of tea cultivars. Few works focus on fine-scale identification across production batches and blended samples of identical tea type, and systematic classification linked to intrinsic fermentation divergence is rarely reported.

The Pearson correlation coefficient quantifies spectral similarity to resolve minor compositional disparities among samples, serving as a powerful tool for batch screening and adulteration identification of highly similar specimens;

this approach has been widely adopted for spectral analysis of agricultural products, minerals and foodstuffs[11]. Pividori et al.[12] developed a Clustermatch correlation-based clustering algorithm capable of capturing both linear and nonlinear correlations to uncover previously unrecognized sex-specific gene expression patterns. Hu et al.[13] put forward the ACCFV algorithm that converts amino acid sequences into feature vectors via positional correlation calculation, realizing 100% classification of 26 HPV subtypes with machine learning. Unfortunately, such refined correlation-based classification approaches are scarcely exploited for high-precision LIBS tea inspection, and no published research combines gradient fermentation differences of tea to correlate classification performance with fermentation quality.

In summary, current LIBS-based tea detection remains fragmented, with separate efforts devoted to either category sorting or quality regression rather than an all-in-one framework incorporating fermentation characteristics. To overcome these limitations, the present work investigates three fermentative grades of tea: non-fermented green tea, semi-fermented oolong tea and post-fermented Pu-erh tea. On the basis of LIBS measurements, Savitzky–Golay (SG) smoothing and background normalization are implemented for spectral preprocessing, followed by variable importance in projection (VIP) feature selection, random forest modeling and Pearson correlation analysis. First, achieve the macroscopic classification of teas with different fermentation properties, as well as the precise identification of samples from the same tea category but from different batches or mixed samples. Subsequently, the screened elemental features are used to build quantitative regression models for core fermentative parameters of Pu-erh tea. This study provides feasible technical support for intelligent non-destructive tea inspection and standardized fermentation optimization.

2. Sample Preparation

2.1 Experimental Facility

The LIBS optical setup adopted in this work features high structural stability and superior detection precision, with its optical configuration sketched in Fig. 1. A Q-switched Nd:YAG solid-state laser operating at 1064 nm was employed, delivering an output energy of 85 mJ at a pulse repetition rate of 5 Hz; the detection integration time was set to 9 μs with zero triggering delay. Spectral acquisition was realized using a four-channel fiber-coupled spectrometer covering an effective wavelength range of 181–835 nm and offering an optical resolution of 0.1 nm.

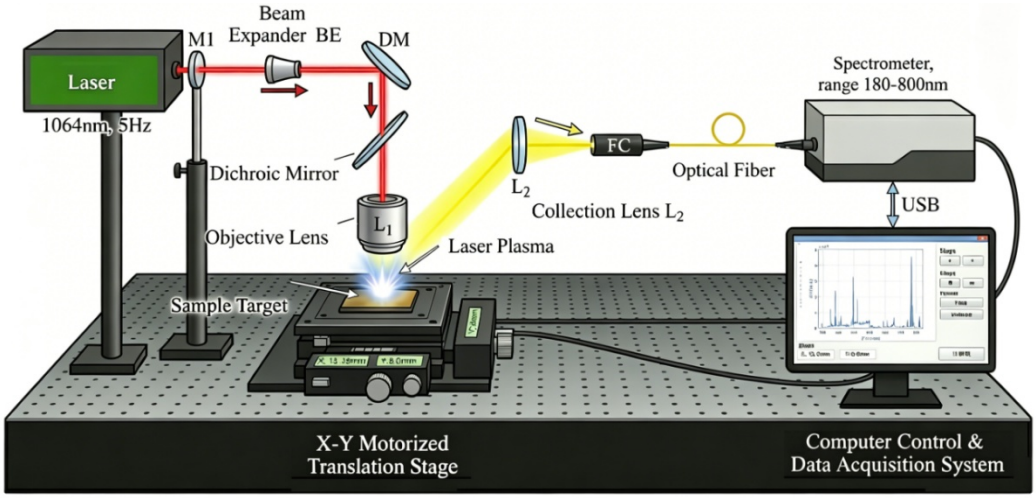


Fig 1 Schematic diagram of the experimental optical path

The incident laser beam was reflected by an optical mirror and subsequently focused onto tea pellets fixed on an XY motorized translation stage. Pulsed laser ablation of the sample surface generates transient plasma. During plasma radiative decay, elemental characteristic emission spectra are produced. An off-axis collection lens captures the plasma emission, which is guided into the four-channel fiber-coupled spectrometer for spectral dispersion and photoelectric conversion to record raw LIBS signals. To suppress experimental errors arising from unstable laser output and irregular sample topography, each specimen is irradiated with 260 replicate laser pulses, guaranteeing the reliability and representativeness of acquired spectra.

Benefiting from integrated sampling, atomization and excitation in a single step, LIBS eliminates cumbersome chemical pretreatment and facilitates synchronous rapid multi-element detection. The peak position, intensity and line profile of characteristic spectra directly reveal elemental speciation and concentration discrepancies in tea, precisely depicting micro-compositional differences induced by cultivar origin, manufacturing batch and fermentation techniques. These spectral fingerprints supply solid microcosmic fundamentals for subsequent tea categorization, fine batch identification and quantitative modeling of fermentation quality.

2.2 Sample Preparation

A graded experimental framework was constructed to hierarchically carry out tea identification and quantitative fermentation characterization. For classification tests, 38 tea specimens were prepared, consisting of 20 semi-fermented oolong teas and 18 non-fermented green teas covering diverse cultivars, production batches, pure raw materials and artificially blended mixtures. These samples comprehensively reflect compositional variations caused by distinct fermentation protocols and manufacturing batches, with detailed sample information listed in Table 1. Typical post-fermented Pu-erh tea was further adopted to establish quantitative models, aiming to reveal the evolving trends of key physicochemical parameters and realize accurate

quantitative evaluation of fermentation quality.

Table 1 Information of Tea Samples

No.	Sample code	Sample difference	Variety	No.	Sample code	Sample difference	Variety
1	A1-1	Same	Jinguanyin	1	B1-1	Same	Large-leaf baked green tea
	A1-2				B1-2		
2	A2-1	Same	Tieguanyin	2	B2-1	Same	Small-medium-leaf baked green tea
	A2-2				B2-2		
3	A3-1	Same	Huangjingui	3	B3-1	Same	Small-medium-leaf pan-fried green tea
	A3-2				B3-2		
4	A4-1	Blended	Tieguanyin	4	B4-1	Blended	Small-medium-leaf baked green tea
	A4-2		Tieguanyin + Benshan		B4-2		Two batches of small-medium-leaf baked green tea
5	A5-1	Blended	Tieguanyin	5	B5-1	Blended	Large-leaf baked green tea
	A5-2		Tieguanyin		B5-2		Two

No.	Sample code	Sample difference	Variety	No.	Sample code	Sample difference	Variety
			+ Benshan				batches of large-leaf baked green tea
6	A6-1	Blended	Jinguanyin	6	B6-1	Blended	Large-leaf baked green tea
	A6-2		Jinguanyin + Benshan		B6-2		Two batches of large-leaf baked green tea
7	A7-1	Blended	Tieguanyin	7	B7-1	Blended	Small-medium-leaf baked green tea
	A7-2		Tieguanyin + Benshan		B7-2		Two batches of small-medium-leaf baked green tea
8	A8-1	Blended	Tieguanyin	8	B8-1	Blended	Large-leaf baked green tea
	A8-2		Tieguanyin + Benshan		B8-2		Large-leaf + small-

No.	Sample code	Sample difference	Variety	No.	Sample code	Sample difference	Variety
							medium-leaf baked green tea
9	A9-1	Blended	Jinguanyin	9	B10-1	Blended	Small-medium-leaf baked green tea
	A9-2		Jinguanyin + Benshan		B10-2		Two batches of small-medium-leaf baked green tea
10	A10-1	Blended	Jinguanyin	—	—	—	—
	A10-2		Jinguanyin + Benshan				

To eliminate matrix effects caused by uneven particle size and surface roughness, and to enhance the stability and repeatability of the LIBS spectral signals, a standardized sample preparation protocol was implemented in this study. All tea samples were thoroughly ground and passed through a 200-mesh standard sieve to remove large particulate impurities. An electronic balance was used to accurately weigh 3 g of tea powder, which was then placed into a tablet press and compressed under a constant pressure of 20 t. This produced standardized circular tea pellets with a diameter of 30 mm and a thickness of 3 mm. Mixed samples were uniformly blended at a 1:1 mass ratio prior to pelletization. The sample preparation workflow is illustrated in Fig. 2.

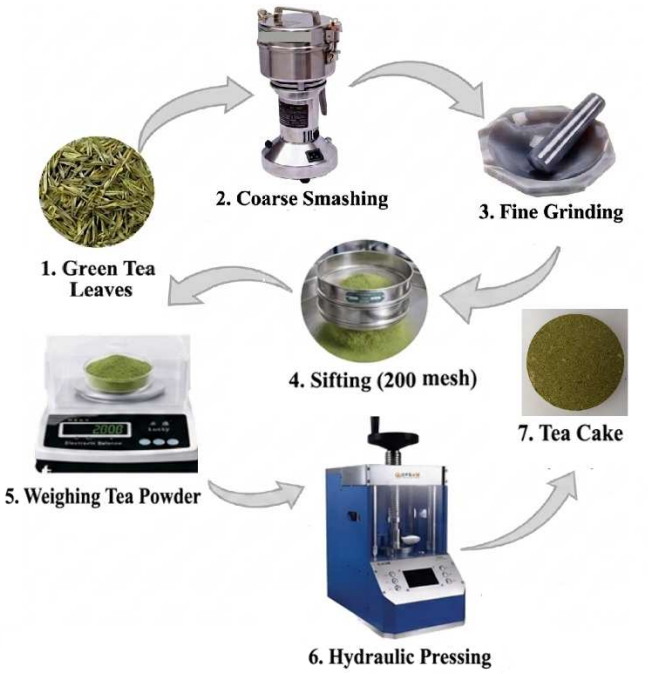


Fig. 2 Sample Preparation Flowchart

All prepared tea pellet samples were sealed and stored in a dark environment to prevent moisture absorption and oxidation, which could compromise experimental accuracy. A representative prepared tea pellet is shown in Fig. 3.



Fig. 3 An individual tea pellet sample

3 Results And Analysis

3.1 Tea Spectral Preprocessing

Raw LIBS spectra are susceptible to laser power fluctuation, instrumental noise, ambient stray light and sample matrix effects, which commonly induce baseline drift and spurious peaks and degrade the accuracy of peak extraction and subsequent modeling. Hence, spectral preprocessing is indispensable. In this experiment, 260 raw spectra were collected per sample. To reduce random noise and improve spectral repeatability, every 52 raw spectra were averaged into a single valid spectrum, yielding a total of 190 effective spectra from 38 tea varieties for further analysis and modeling.

Several preprocessing strategies were comparatively evaluated, and the combination of Savitzky–Golay (SG) smoothing and background normalization was finally selected. Conventional interpolation tends to discard critical spectral features and distort peak intensity and profile during background

subtraction; moving-average smoothing may truncate genuine peaks , resulting in signal distortion[14]. By contrast, the SG algorithm fits spectral profiles via the least-squares criterion, which suppresses high-frequency noise and corrects baseline drift while fully preserving the position, intensity and shape of characteristic peaks. Such treatment retains the intrinsic elemental spectral signatures of tea as much as possible, and the spectral comparison of oolong tea before and after preprocessing is illustrated in Fig. 4.

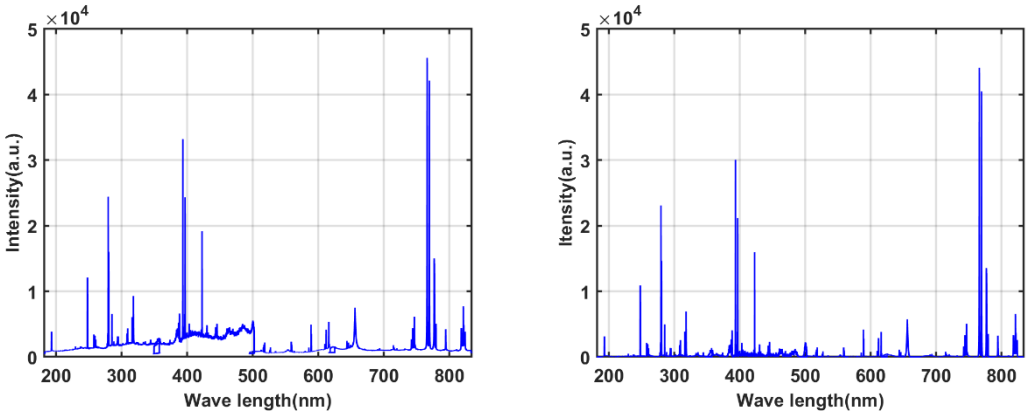


Fig.4 LIBS Spectral Diagram of Oolong Tea
(a) Original Spectrum (b) Spectrum after Removing Background

Preprocessed LIBS spectra were assigned to specific elemental emission lines by referencing the atomic spectral database from the National Institute of Standards and Technology (NIST). Characteristic peaks originating from mineral elements Mn, Mg, Al, Ca, Fe, K and nonmetallic elements C, O, N were clearly resolved, as presented in Fig. 5.

These elemental signatures are strongly correlated with tea cultivar, cultivation environment and fermentation procedure. The accumulation of mineral elements is governed by tea plant genotype and soil uptake behavior, serving as microscale fingerprints for cultivar discrimination. Nonmetallic species including C and N originate from intrinsic organic components such as tea polyphenols and carbohydrates, whose spectral intensities directly quantify the degradation and transformation of organics during fermentation. These features supply characteristic markers for fermentative-type classification and lay a physical foundation for subsequent quantitative modeling of fermentation quality.

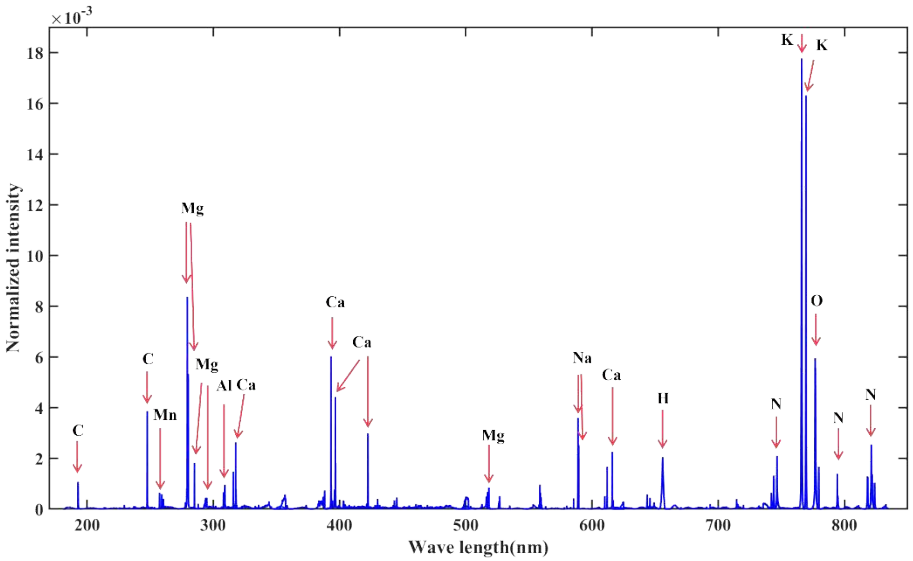


Fig.5 LIBS spectrum of tea

3.2 Random Forest Modeling Based On VIP Feature Selection

Random forest is an ensemble machine-learning algorithm composed of multiple decision trees. It exhibits prominent capacity for high-dimensional data processing and nonlinear fitting, effectively alleviating overfitting risks of individual models with favorable stability and generalization, which renders it highly suitable for modeling high-dimensional LIBS spectral datasets[15]. To eliminate redundant spectral variables and screen core features closely linked to tea variety and fermentation quality, the variable importance in projection (VIP) method was adopted for feature optimization, pinpointing elemental signatures sensitive to varietal divergence and graded fermentation variations to construct robust input parameters.

Five alternative machine-learning algorithms, namely K-nearest neighbor (KNN), Naive Bayes, Linear Discriminant Analysis (LDA), Decision Tree and Artificial Neural Network (ANN), were established for comparative classification between green tea and oolong tea. Experimental results in Fig. 6 reveal that RF achieves the optimal performance with a 100% classification accuracy for coarse tea-type discrimination. Nevertheless, standalone machine-learning models fail to distinguish subtle discrepancies among highly similar samples from identical tea categories, such as different production batches. To address this limitation, the Pearson correlation coefficient was introduced to quantify spectral similarity and enable refined sample identification. Teas from identical cultivar and batch share nearly consistent elemental and organic compositions and hence possess remarkably similar spectra; by contrast, deviations in fermentation conditions and component proportion induce quantifiable spectral differences across distinct batches and blended samples, which can be precisely captured via correlation calculation to realize fine-grained classification.

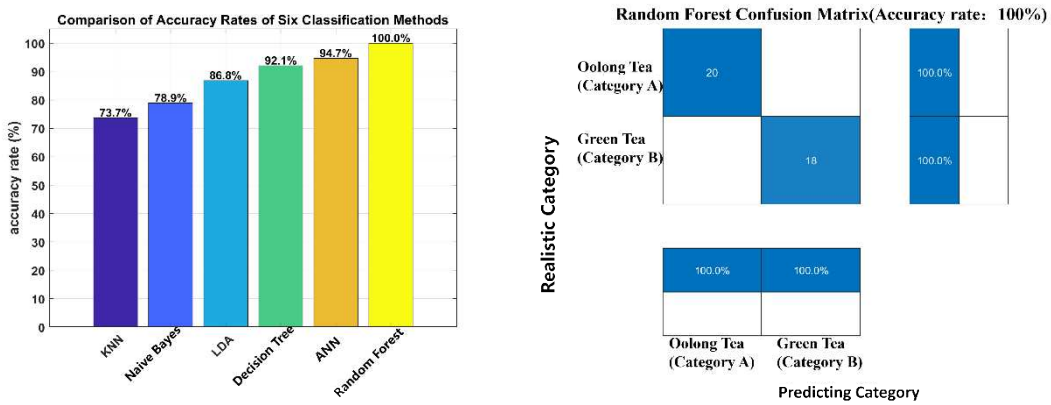


Fig 6 Classification of Tea Categories (a) Comparison chart of accuracy rates for six classification methods (b) Confusion matrix of Random Forest classification

VIP based weighting screening demonstrates that discriminative identification of teas with distinct cultivars and fermentation extents relies on synergistic responses of specific elemental emission lines, consistent with previously reported findings[16-17]. Redundant and low-contribution spectral variables were iteratively removed via VIP ranking, and nine characteristic elemental lines. C, Ca, Na, K, Fe, Al, Mg, Mn, Si were finally screened for semi-fermented oolong tea identification, as displayed in Fig. 7(a).

Existing literatures verify that enzymatic oxidation during oolong semi-fermentation remarkably rearranges organic structures and modulates mineral accumulation. The enrichment patterns of Mg, Mn and Al are genotype-dependent and serve as critical spectral fingerprints to differentiate oolong varieties[16]. Specifically, Mg and Mn participate in photosynthesis and aroma biosynthesis of tea plants; inter-variety metabolic discrepancies produce distinguishable spectral intensities for Tieguanyin, Jinguanyin and Huangjingui. Variation in Al content further enables germplasm discrimination between Benshan tea and common oolong cultivars, facilitating adulteration detection. In addition, carbon-related organic emission lines directly trace oxidative degradation induced by semi-fermentation and characterize processing-derived compositional variations[17]. Such results confirm that VIP selected elemental features are theoretically sound and competent for oolong tea classification.

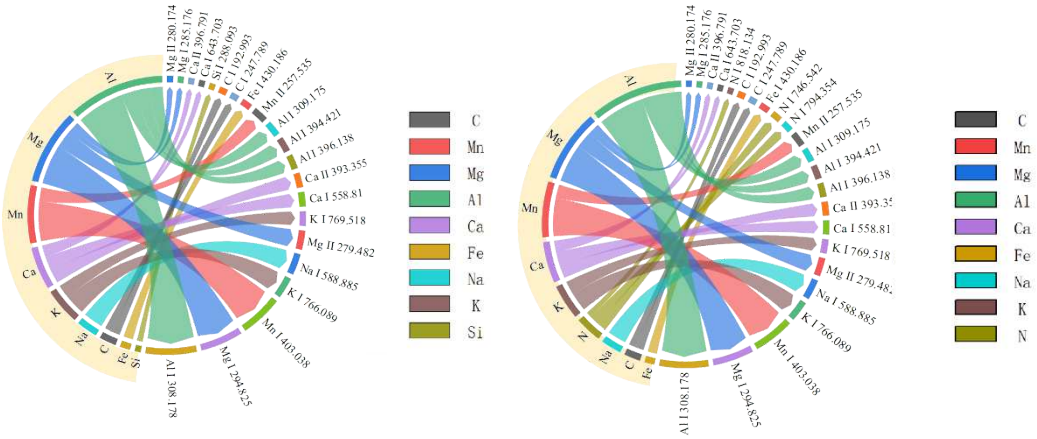


Fig 7 VIP Feature Element Screening (a) Type A Oolong Tea (b) Type B Green Tea

For non-fermented green tea specimens, nine high-correlation elemental emission lines. C, N, Ca, Na, K, Fe, Al, Mg, Mn were screened via VIP optimization, as illustrated in Fig. 7(b). This selection agrees well with published research on mineral-fingerprint-based green tea identification[18-19]. Owing to the absence of fermentative oxidation, varietal and processing disparities of green tea mainly originate from photosynthetic mineral accumulation and thermally induced organic transformation. Stable enrichment of Mg, Mn, K and Al provides fundamental markers for germplasm tracing and variety classification of green tea[18]. The screened spectral features conform to such rules: Mg and Mn lines discriminate photosynthetic divergence between large-leaf and small-leaf tea cultivars; Ca-related peaks are highly sensitive to thermal manufacturing including high-temperature pan-firing and low-temperature oven-drying, enabling precise identification of processing differences; C and N organic signals directly reflect organic content and thermal carbonization degree, comprehensively covering cultivar and processing variations of green tea[19].

Combined with published literature, the VIP selected characteristic lines for green and oolong teas well match their respective fermentation mechanisms, germplasm properties and processing discrepancies. These features support fine classification of tea cultivars, production batches and blended samples while sensitively reflecting intrinsic fermentation variation, laying a solid feature foundation for subsequent fermentation quality modeling.

3.3 Analysis Of Classification Results

Using VIP optimized elemental features, random forest and threshold-based Pearson correlation analysis were combined to realize coarse tea grouping and refined batch discrimination. A threshold of 0.999 was predefined for spectral similarity: samples with correlation coefficients above this value were regarded as identical batches, whereas values below the threshold indicated distinct batches or blended mixtures.

Correlation matrices for oolong and green tea batches are summarized in Fig. 8 and Table 2, Fig. 9 and Table 3, respectively. All Pearson coefficients between different batches of identical pure tea exceed 0.999, indicating highly consistent elemental composition and fermentation quality. In contrast, coefficients from cross-batch or blended samples fall below the set threshold with obvious spectral discrepancies. Benefiting from the integrated strategy of VIP feature selection, random forest modeling and Pearson threshold judgment, 100% accuracy was achieved for both coarse categorization of green, oolong tea and refined identification of batches and blended samples within each tea type.

Varied fermentation degrees alter the content of tea polyphenols and organic compounds, generating distinguishable elemental spectral responses for macro classification between green and oolong tea. Meanwhile, subtle discrepancies from batch fluctuation and blending proportion can be captured by VIP screened sensitive elements and further quantified via Pearson correlation, which strongly validates the excellent adaptability of selected spectral features to cultivar and fermentation variations.

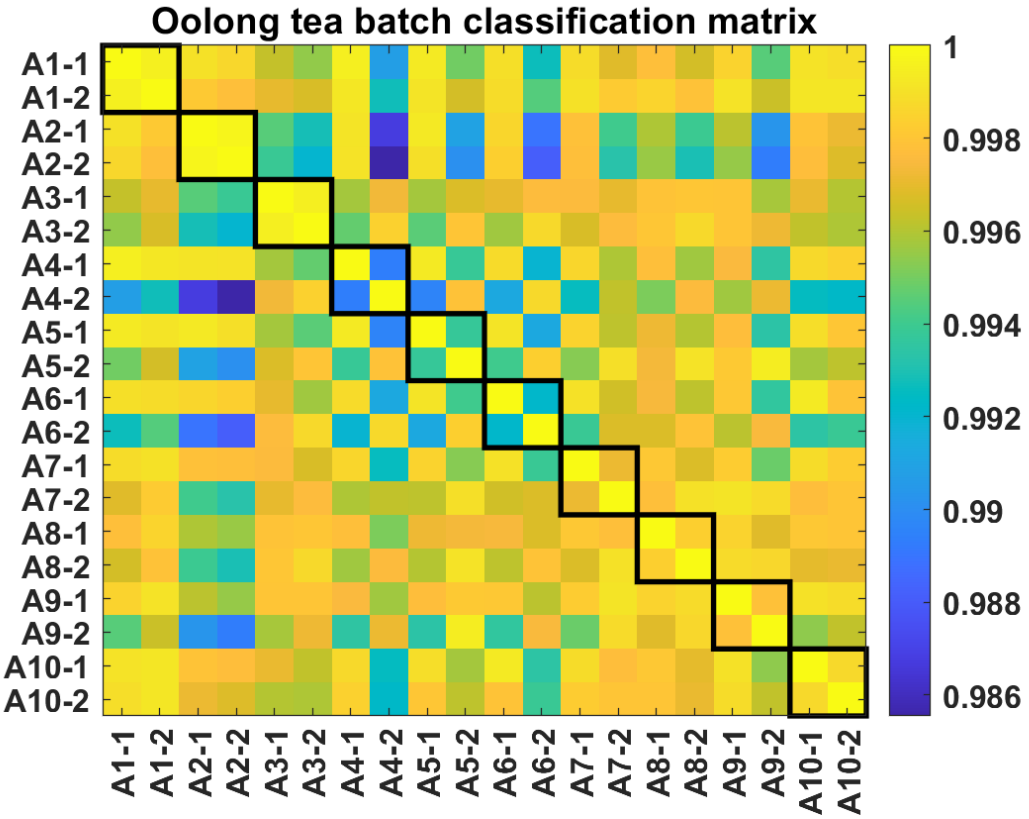


Fig 8 Batch Classification Matrix for Oolong Tea

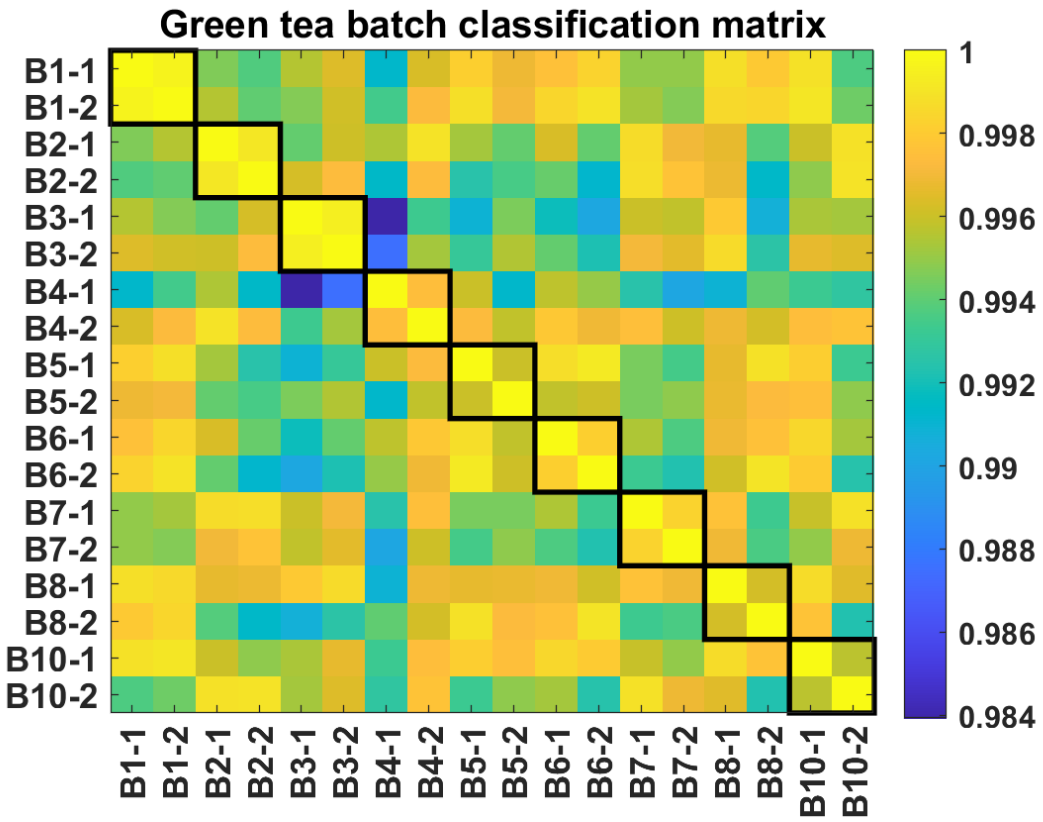


Fig 9 Batch Classification Matrix for Green Tea

Table 2 Pearson Correlation Coefficient of Oolong Tea

	A1-1	A1-2	A2-1	A2-2	A3-1	A3-2	A4-1	A4-2	A5-1	A5-2	A6-1	A6-2	A7-1	A7-2	A8-1	A8-2	A9-1	A9-2	A10-1	A10-2
A1-1	1.0000	0.9996	0.9990	0.9987	0.9963	0.9954	0.9995	0.9908	0.9993	0.9949	0.9989	0.9927	0.9989	0.9969	0.9978	0.9966	0.9985	0.9945	0.9991	0.9989
A1-2	0.9996	1.0000	0.9982	0.9978	0.9970	0.9967	0.9993	0.9927	0.9991	0.9966	0.9989	0.9945	0.9990	0.9982	0.9986	0.9979	0.9991	0.9964	0.9992	0.9992
A2-1	0.9990	0.9982	1.0000	0.9998	0.9945	0.9929	0.9991	0.9867	0.9993	0.9910	0.9986	0.9890	0.9978	0.9940	0.9959	0.9939	0.9961	0.9903	0.9980	0.9971
A2-2	0.9987	0.9978	0.9998	1.0000	0.9938	0.9920	0.9990	0.9856	0.9990	0.9901	0.9984	0.9881	0.9978	0.9933	0.9956	0.9929	0.9955	0.9893	0.9977	0.9968
A3-1	0.9963	0.9970	0.9945	0.9938	1.0000	0.9995	0.9957	0.9973	0.9957	0.9968	0.9970	0.9976	0.9976	0.9970	0.9979	0.9981	0.9980	0.9958	0.9971	0.9960
A3-2	0.9954	0.9967	0.9929	0.9920	0.9995	1.0000	0.9947	0.9984	0.9946	0.9980	0.9957	0.9988	0.9967	0.9977	0.9980	0.9987	0.9980	0.9972	0.9962	0.9959
A4-1	0.9995	0.9993	0.9991	0.9990	0.9957	0.9947	1.0000	0.9894	0.9993	0.9938	0.9988	0.9919	0.9986	0.9959	0.9978	0.9957	0.9975	0.9935	0.9987	0.9985
A4-2	0.9908	0.9927	0.9867	0.9856	0.9973	0.9984	0.9894	1.0000	0.9896	0.9979	0.9913	0.9987	0.9926	0.9963	0.9951	0.9976	0.9957	0.9971	0.9925	0.9923
A5-1	0.9993	0.9991	0.9993	0.9990	0.9957	0.9946	0.9993	0.9896	1.0000	0.9938	0.9991	0.9913	0.9985	0.9962	0.9972	0.9960	0.9977	0.9934	0.9989	0.9980
A5-2	0.9949	0.9966	0.9910	0.9901	0.9968	0.9980	0.9938	0.9979	0.9938	1.0000	0.9940	0.9984	0.9953	0.9990	0.9974	0.9991	0.9982	0.9994	0.9957	0.9962
A6-1	0.9989	0.9989	0.9986	0.9984	0.9970	0.9957	0.9988	0.9913	0.9991	0.9940	1.0000	0.9922	0.9990	0.9965	0.9974	0.9962	0.9981	0.9936	0.9993	0.9980
	0.9927	0.9945	0.9890	0.9881	0.9976	0.9988	0.9919	0.9987	0.9913	0.9984	0.9922									

A6-2	0.9927	0.9945	0.9890	0.9881	0.9976	0.9988	0.9919	0.9987	0.9913	0.9984	0.9922	1.0000	0.9939	0.9968	0.9968	0.9979	0.9961	0.9975	0.9934	0.9939
A7-1	0.9989	0.9990	0.9978	0.9978	0.9976	0.9967	0.9986	0.9926	0.9985	0.9953	0.9990	0.9939	1.0000	0.9971	0.9981	0.9967	0.9983	0.9948	0.9989	0.9983
A7-2	0.9969	0.9982	0.9940	0.9933	0.9970	0.9977	0.9959	0.9963	0.9962	0.9990	0.9965	0.9968	0.9971	1.0000	0.9978	0.9990	0.9991	0.9988	0.9977	0.9980
A8-1	0.9978	0.9986	0.9959	0.9956	0.9979	0.9980	0.9978	0.9951	0.9972	0.9974	0.9974	0.9968	0.9981	0.9978	1.0000	0.9984	0.9985	0.9969	0.9981	0.9980
A8-2	0.9966	0.9979	0.9939	0.9929	0.9981	0.9987	0.9957	0.9976	0.9960	0.9991	0.9962	0.9979	0.9967	0.9990	0.9984	1.0000	0.9988	0.9987	0.9970	0.9971
A9-1	0.9985	0.9991	0.9961	0.9955	0.9980	0.9980	0.9975	0.9957	0.9977	0.9982	0.9981	0.9961	0.9983	0.9991	0.9985	0.9988	1.0000	0.9978	0.9990	0.9988
A9-2	0.9945	0.9964	0.9903	0.9893	0.9958	0.9972	0.9935	0.9971	0.9934	0.9994	0.9936	0.9975	0.9948	0.9988	0.9969	0.9987	0.9978	1.0000	0.9954	0.9962
A10-1	0.9991	0.9992	0.9980	0.9977	0.9971	0.9962	0.9987	0.9925	0.9989	0.9957	0.9993	0.9934	0.9989	0.9977	0.9981	0.9970	0.9990	0.9954	1.0000	0.9987
A10-2	0.9989	0.9992	0.9971	0.9968	0.9960	0.9959	0.9985	0.9923	0.9980	0.9962	0.9980	0.9939	0.9983	0.9980	0.9980	0.9971	0.9988	0.9962	0.9987	1.0000

Table 3 Pearson Correlation Coefficient of Green Tea

	B1-1	B1-2	B2-1	B2-2	B3-1	B3-2	B4-1	B4-2	B5-1	B5-2	B6-1	B6-2	B7-1	B7-2	B8-1	B8-2	B10-1	B10-2
B1-1	1.0000	0.9997	0.9946	0.9937	0.9956	0.9965	0.9913	0.9963	0.9982	0.9969	0.9976	0.9983	0.9949	0.9949	0.9989	0.9979	0.9989	0.9937

B1-2	0.999 7	1.000 0	0.995 5	0.994 1	0.994 7	0.996 1	0.993 4	0.997 3	0.998 9	0.997 1	0.998 5	0.999 0	0.995 2	0.994 7	0.998 6	0.998 5	0.999 1	0.994 3
B2-1	0.994 6	0.995 5	1.000 0	0.999 1	0.994 1	0.996 0	0.995 5	0.999 0	0.995 3	0.994 1	0.996 3	0.994 1	0.998 7	0.997 0	0.996 7	0.993 8	0.996 0	0.998 9
B2-2	0.993 7	0.994 1	0.999 1	1.000 0	0.996 3	0.997 4	0.991 4	0.997 4	0.992 5	0.993 6	0.994 2	0.991 2	0.998 8	0.997 7	0.996 8	0.991 4	0.994 9	0.999 0
B3-1	0.995 6	0.994 7	0.994 1	0.996 3	1.000 0	0.999 4	0.983 9	0.993 3	0.990 9	0.994 5	0.991 9	0.990 3	0.996 0	0.995 8	0.997 9	0.990 7	0.995 4	0.995 2
B3-2	0.996 5	0.996 1	0.996 0	0.997 4	0.999 4	1.000 0	0.987 5	0.995 3	0.993 1	0.995 5	0.994 2	0.992 2	0.997 1	0.996 6	0.998 7	0.992 6	0.996 7	0.996 4
B4-1	0.991 3	0.993 4	0.995 5	0.991 4	0.983 9	0.987 5	1.000 0	0.997 4	0.996 0	0.991 3	0.995 8	0.995 0	0.992 5	0.990 1	0.990 9	0.994 1	0.993 3	0.992 8
B4-2	0.996 3	0.997 3	0.999 0	0.997 4	0.993 3	0.995 3	0.997 4	1.000 0	0.997 3	0.995 8	0.997 9	0.996 9	0.997 5	0.996 1	0.996 9	0.996 2	0.997 4	0.997 7
B5-1	0.998 2	0.998 9	0.995 3	0.992 5	0.990 9	0.993 1	0.996 0	0.997 3	1.000 0	0.996 0	0.998 8	0.999 3	0.994 5	0.993 5	0.996 7	0.998 9	0.998 1	0.993 2
B5-2	0.996 9	0.997 1	0.994 1	0.993 6	0.994 5	0.995 5	0.991 3	0.995 8	0.996 0	1.000 0	0.995 8	0.996 1	0.994 5	0.994 9	0.996 7	0.997 3	0.997 5	0.994 9
B6-1	0.997 6	0.998 5	0.996 3	0.994 2	0.991 9	0.994 2	0.995 8	0.997 9	0.998 8	0.995 8	1.000 0	0.998 2	0.995 4	0.993 7	0.997 0	0.997 6	0.998 5	0.995 3
B6-2	0.998 3	0.999 0	0.994 1	0.991 2	0.990 3	0.992 2	0.995 0	0.996 9	0.999 3	0.996 1	0.998 2	1.000 0	0.993 2	0.992 3	0.996 2	0.999 0	0.998 0	0.992 4
B7-1	0.994 9	0.995 2	0.998 7	0.998 8	0.996 0	0.997 1	0.992 5	0.997 5	0.994 5	0.994 5	0.995 4	0.993 2	1.000 0	0.998 4	0.997 6	0.993 3	0.995 9	0.998 9

B7-2	0.9949	0.9947	0.9970	0.9977	0.9958	0.9966	0.9901	0.9961	0.9935	0.9949	0.9937	0.9923	0.9984	1.0000	0.9969	0.9936	0.9949	0.9969
B8-1	0.9989	0.9986	0.9967	0.9968	0.9979	0.9987	0.9909	0.9969	0.9967	0.9967	0.9970	0.9962	0.9976	0.9969	1.0000	0.9962	0.9987	0.9965
B8-2	0.9979	0.9985	0.9938	0.9914	0.9907	0.9926	0.9941	0.9962	0.9989	0.9973	0.9976	0.9990	0.9933	0.9936	0.9962	1.0000	0.9977	0.9923
B10-1	0.9989	0.9991	0.9960	0.9949	0.9954	0.9967	0.9933	0.9974	0.9981	0.9975	0.9985	0.9980	0.9959	0.9949	0.9987	0.9977	1.0000	0.9957
B10-2	0.9937	0.9943	0.9989	0.9990	0.9952	0.9964	0.9928	0.9977	0.9932	0.9949	0.9953	0.9924	0.9989	0.9969	0.9965	0.9923	0.9957	1.0000

3.4 Fermentation Quality Analysis of Pu-erh Tea

Utilizing the same set of fermentation-sensitive core characteristic elements previously selected via the VIP algorithm, fermentation quality modeling was conducted specifically for post-fermented Pu-erh tea. As a typical post-fermented tea, Pu-erh undergoes systematic changes in moisture, water extracts, and tea polyphenol contents during the fermentation process. These are the core indicators that characterize its fermentation degree and overall quality.

The underlying modeling mechanism indicates that during Pu-erh tea fermentation, microbial activity drives the gradual degradation of tea polyphenols, the transformation of organic matter, and dynamic fluctuations in moisture content. The alteration of these internal constituents directly impacts the spectral response intensities of core characteristic elements in the tea, such as C, Mg, Mn, and Ca.

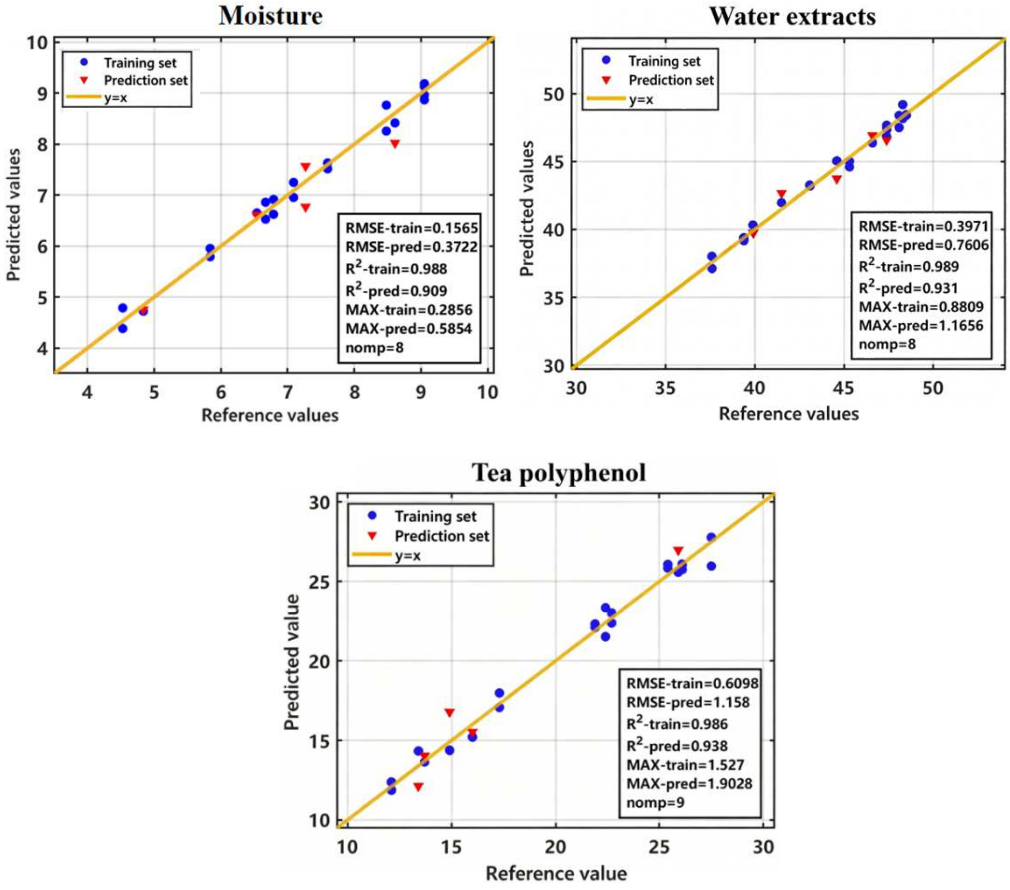


Fig. 10 Prediction models for moisture, water extracts, and tea polyphenols in Pu-erh tea.

As shown in Fig. 10, by constructing a Random Forest quantitative model based on the uniformly optimized characteristic elements, the nonlinear correlations between the characteristic spectral signals and the three major fermentation indicators can be precisely fitted. This capability enables both the qualitative discrimination and quantitative prediction of the fermentation degree in Pu-erh tea, thereby refining the theoretical framework for the

spectral detection of tea fermentation.

4 Conclusions

Focusing on fermentation differences as the core criterion, this study investigated three typical tea categories: non-fermented green tea, semi-fermented oolong tea, and post-fermented Pu-erh tea. Based on LIBS technology coupled with VIP feature selection, the Random Forest algorithm, and Pearson correlation coefficient analysis, systematic research was conducted on three main aspects: cross-category fermentation classification, fine-grained discrimination of fermentation batches within the same category, and quantitative modeling of fermentation quality in post-fermented tea. This study elucidates the spectral response patterns of teas subjected to different fermentation mechanisms and perfects the tea fermentation classification system based on optical spectroscopy. By overcoming the subjective limitations of traditional experience-based sensory evaluation, this research reveals the differential response behaviors of various fermentation processes at a microscopic spectral level. It provides novel technical methodologies and theoretical support for fermentation process traceability, standardized fermentation grading, mixed sample authentication, and the intelligent control of fermentation quality, demonstrating significant application value in the fields of intelligent tea processing and optical non-destructive testing.

Author contributions

Jiaqi Zhao: investigation, methodology, data curation, writing original draft. Yongyi Du: validation. Yue Lv: validation. Shuoyu Yang: validation. Ruibin Liu: conceptualization, writing – review & editing, project administration.

Conflicts of interest

There are no conflicts to declare.

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Data availability

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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